

Pilates-based therapeutic exercise for pregnancy-related low back and pelvic pain: A prospective, randomized, controlled trial.

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In this study, we aimed to clarify the impact of a Pilates-based therapeutic exercise on disability, pain, mood, and sleep quality in patients with pregnancy-related lumbopelvic pain (LPP). In the single-blinded randomized controlled study conducted between January 2018 and June 2018, 34 pregnant women (mean age: 29.7 ± 6.2 years; range, 18 to 40 years) in the second trimester (week 14-24) with LPP were randomly assigned to a control group and a Pilates group. All patients underwent usual prenatal care. In addition, the selected Pilates exercise was carried out twice a week for 60-min per session for 12 weeks in the Pilates group. The control group was not prescribed an exercise regimen; however, they were not discouraged from exercising. The primary outcome was disability; secondary outcomes were LPP, mood, and sleep quality. Disability [Roland-Morris Disability Questionnaire (RMDQ)], LPP [Visual Analog Scale (VAS)], mood [Hospital Anxiety and Depression Scale, Anxiety (HADS-A) and Depression (HADS-D) subscales], and sleep quality [Pittsburgh Sleep Quality Index (PSQI)] were measured before and after 12 weeks. Adverse effects and adherence were recorded to determine exercise safety and compliance. The intention-to-treat analysis was applied. The between-group effect sizes were moderate for the RMDQ and VAS scales ($d=0.4$ and $d=0.7$, respectively) and small for the HADS-A and HADS-D scales (both $d=0.2$). The intention-to-treat analysis demonstrated that there was a statistically significant difference in disability, pain, and mood in favor of the Pilates group ($p<0.05$). Adding Pilates to usual prenatal care should be considered a promising treatment option for pregnancy-related LPP.

Effects of pilates training on lumbo-pelvic stability and flexibility.

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This study was performed to assess and compare the effects of Pilates exercise on flexibility and lumbo-pelvic movement control between the Pilates training and control groups. A randomized single-blinded controlled design was utilized in the study. Forty healthy male and female volunteers (mean age 31.65 ± 6.21 years) were randomly divided into Pilates-based training (20 subjects) and the control groups (20 subjects). The Pilates group attended 45-minute training sessions, 2 times per week, for a period of 8 weeks. Flexibility and lumbo-pelvic stability tests were determined as outcome measures using a standard "sit and reach test" and "pressure biofeedback" respectively at 0, 4 and 8 weeks of the study. The results showed that the Pilates training group improved flexibility significantly ($P<0.001$) during time intervals. This effect was also significantly greater than the control group for both 4 weeks and 8 weeks of the training period ($P<0.001$). There were 65% and 85% of the subjects from Pilates group passing the lumbo-pelvic stability test at 4 and 8 weeks of training periods respectively. No subjects from the control group passed the test at any stages. Pilates can be used as an adjunctive exercise program to improve flexibility, enhance control-mobility of trunk and pelvic segments. It may also prevent and attenuate the predisposition to axial musculoskeletal injury.

Does pilates-based exercise improve postural alignment in adult women?

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A randomized, controlled trial was conducted to determine the effect of Pilates-based exercise on postural alignment. Seventy-four adult women (mean age $\pm$ SD, 34.9 ± 16.4 years) were randomized to a Pilates-based mat class ($n = 40$) or a control group ($n = 34$). Pilates-based exercise participants were taught the Initial Mat of Body Control Pilates for 6 months, twice a week, for 60 minutes per session; the control group received no exercise intervention. Repeated measurements were performed at baseline, 3 months, and 6 months of the frontal alignment of the thoracolumbar spine, shoulder, and pelvis, and sagittal alignment of the head and pelvis. No differences were found in either group, over time, on frontal alignment of the thoracolumbar spine and pelvis. The experimental group showed significant improvements in frontal alignment of the shoulder and sagittal alignment of the head and pelvis at 6 months. The Pilates-based exercise enhanced some parameters of the postural alignment of women, as measured by frontal alignment of the shoulder and sagittal alignment of the head and pelvis. The significant improvement in sagittal alignment of the head may imply that 6 months of Pilates-based exercise enhances sagittal alignment of the cervical or thoracic spine.

[THE EFFECTS OF THE PILATES METHOD ON HAMSTRING EXTENSIBILITY, PELVIC TILT AND TRUNK FLEXION].

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Pilates includes a high volume of hamstring stretching and maximal trunk flexion with knees extended exercises. to perform a systematic review about Pilates practice effects and a detraining period on hamstring extensibility, pelvic tilt and trunk flexion in maximal trunk flexion with knees extended. it was analysed all the experimental or quasi-experimental designs written in English, Spanish or Portuguese and included in the following databases: Pubmed, Sports Discus, ISI Web of Knowledge, Dialnet and Research Gate. twenty-one papers were analysed. Most of them used a pre-test-post-test design with control group. The intervention programs applied were heterogeneous. Samples were composed mainly of women, both young and old. It was found that the Pilates practice, with different volume, significantly increased hamstring muscle extensibility and pelvic tilt in maximal trunk flexion. At least three training sessions per week during six weeks were necessary in order to obtain a high trunk inclination. Studies which involved athletes showed contradictory results. By inducing a detraining period it was noticed a decrease in hamstring extensibility and trunk flexion from the second week. there is a moderate evidence that Pilates is an effective method to increase hamstring extensibility, pelvic tilt and the degree of trunk flexion in maximal flexion positions in sedentary and recreational active people and also to increase hamstring extensibility in athletes.

Effect of Home-Based Tele-Pilates Intervention on Pregnant Women: A Pilot Study.

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Pilates is effective for training the core muscles and stabilizing the hip joints, which provides relief from pelvic pain and low back pain during pregnancy. However, there are no specific guidelines on appropriate physical exercises for pregnant women due to the current pandemic. We aimed to apply the exercise standard proposed by the American College of Obstetricians and Gynecologists to home-based tele-Pilates exercise (HTPE), to determine its effect on the physical and mental health of pregnant women. We randomly divided the subjects into the following two groups who completed 8 weeks of HTPE

(50 min/day, 2 days/week): (a) Pilates exercise (PE, $n = 7$) and (B) non-Pilates exercise (CON, $n = 7$). HTPE was performed by adjusting the program every 3 weeks, based on pain and physical fitness levels. We measured body composition, muscles of the hip joint, pelvic tilt, Oswestry Disability Index (ODI), and Pittsburgh Sleep Quality Index (PSQI), before and after HTPE. Following HTPE, while the percentage of body fat and body mass index had significantly decreased, the body fat mass did not change in the PE group ($p < 0.05$). The PE group showed an increase in strength of the left and right hip flexion and hip abduction, compared to the CON group ($p < 0.01$). The ODI and PSQI were significantly decreased in the PE group ($p < 0.05$). Therefore, the 8-week HTPE program is an effective exercise for pregnant woman that reduces body fat metabolism and strengthens muscles of the hip joint, thus alleviating pregnancy-induced low back pain and insomnia.

Obstructive sleep apnoea-associated factors in children and adolescents diagnosed by polysomnography: A scoping review.

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This scoping review aimed to map research on factors associated with obstructive sleep apnoea (OSA) in children and adolescents undergoing overnight polysomnography (PSG) and questionnaire-based diagnostic assessments. Searches were conducted in three electronic databases up to May 2023, including nine observational studies, including 3482 individuals. Among the included studies, nine reported on sex, six on obesity, five on tonsillar hypertrophy, three on mouth breathing, two on allergic rhinitis, and three on smoking exposure. The sample comprised 3482 children, with subsets analyzed for sex ($n = 3482$), obesity ($n = 2752$), and tonsillar hypertrophy ($n = 794$). Meta-analysis demonstrated a significantly higher prevalence of OSA in males compared to females [$P < 0.0001$; $I^2 = 49\%$], with a pooled relative risk (RR) of 1.15 (95 % confidence interval [CI]: 1.07-1.23). Associations were found between obesity and OSA [RR: 1.42; 95 % CI: 1.20-1.68; $P = 0.02$; $I^2 = 61\%$], and tonsillar size and OSA [RR: 1.61; 95 % CI: 1.35-1.92; $P = 0.06$; $I^2 = 60\%$]. Considering the study's limitations, these findings underscore the importance of considering sex, obesity, and tonsillar size when evaluating OSA in children and adolescents.

Sleep apnea is a common and dangerous cardiovascular risk factor.

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Sleep apnea involves almost one billion individuals throughout the world, including 40 million Americans. Of major medical concern is the fact that the prevalence of sleep apnea is significantly increasing due to the epidemic of obesity, physical inactivity, and diabetes mellitus which are important risk factors for the development and persistence of sleep apnea in individuals. Sleep apnea is characterized by multiple episodes of apnea or hypopnea during sleep, which cause nocturnal arousals, gasping for breath during the night, daytime sleepiness, irritability, forgetfulness, fatigue and recurrent headaches. Obstructive sleep apnea occurs when upper airway obstruction occurs in an individual during sleep with absent or markedly reduced airflow in the presence of continued activity of inspiratory thoracic and diaphragmatic muscles. Central sleep apnea is defined as the absence or the significant reduction of naso-oral airflow due to the withdrawal during sleep of ponto-medullary respiratory center stimulation of the nerves of the inspiratory thoracic and diaphragmatic muscles and absence of contraction of these muscles during

apnea. Complex sleep apnea occurs when an individual exhibits characteristics of both obstructive and central sleep apnea. The severity of sleep apnea is measured by polysomnography and the apnea hypopnea index (AHI), which is the average number of apneas and hypopneas per hour of sleep measured by polysomnography. Sleep apnea is mild if the AHI is 5-14/h with no or mild symptoms, moderate if the AHI is 15 to 30/h with occasional daytime sleepiness, and severe if the AHI is >30/h with frequent daytime sleepiness that interferes with the normal activities of daily life. Chronic sleep apneas and hypopneas followed by compensatory hyperpneas are associated with significant adverse cardiovascular consequences including: 1) recurrent hypoxemia and hypercarbia; 2) Increased sympathetic nerve activity and decreased parasympathetic nerve activity; 3) oxidative stress and vascular endothelial dysfunction; and 4) cardiac remodeling and cardiovascular disease. Moderate or severe sleep apnea significantly increases the risk of coronary artery disease, congestive heart failure, cerebral vascular events (strokes), and cardiac dysrhythmias, and also increase the morbidity and mortality of these diseases. Nevertheless, sleep apnea is currently underdiagnosed and untreated in many individuals due to the challenges in the prediction and detection of sleep apnea and a lack of well-defined optimal treatment guidelines. Chronic continuous positive airway pressure for ≥ 4 h/night for >70% of nights is beneficial in the treatment of patients with sleep apnea. CPAP Improves sleep quality, reduces the AHI, augments cardiac output and increases oxygen delivery to brain and heart, reduces resistant hypertension, decreases cardiac dysrhythmias, and reduces daytime sleepiness. The present article discusses the diagnosis of obstructive sleep apnea, central sleep apnea, and complex apnea. Thereafter the important pathophysiologic mechanisms in sleep apnea and the relationship of these pathophysiologic mechanics to atherosclerotic vascular disease are reviewed. Guidelines are then provided for the treatment of mild, moderate and severe sleep apnea. In order to reduce the cardiovascular morbidity and mortality caused by sleep apnea and facilitate the diagnosis and the long-term, effective treatment of sleep apnea in patients, the close cooperation is necessary of cardiovascular specialists, pulmonary specialists, and respiratory therapy/rehabilitation specialists.

Polysomnographic features of children with obesity: body mass index predict severe obstructive sleep apnea in obese children?

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Few studies have explored the polysomnographic features of children with obesity. This study aimed to explore the demographic and polysomnographic features of obese children and determine whether body mass index (BMI) could predict severe obstructive sleep apnea (OSA). This cross-sectional study recruited obese children who underwent diagnostic polysomnography between January 2019 and March 2022. We explored demographic and anthropometric measures as well as polysomnographic abnormalities among them. We used receiver operating characteristic curves and logistic regression analyses to determine the optimal cutoff values of anthropometric variables for predicting severe OSA. A total of 132 children with obesity (76.5% male; mean age, 12.5 ± 3.2 years) were included. Severe OSA was identified in 64 children (48.5%). Desaturation was observed in 59.8%, while 23.5% had hyperarousal, 20.5% had sleep-related hypoventilation, 60.6% had positional OSA, 40.2% had rapid eye movement-related OSA, and 5.0% had obesity hypoventilation syndrome. Among them, BMI (odds ratio [OR], 1.11; 95% confidence interval [CI], 1.05-1.17; $P < 0.001$), neck circumference (OR, 1.15; 95% CI, 1.07-1.25; $P < 0.001$), and waist circumference (OR, 1.04; 95% CI, 1.02- 1.07; $P = 0.001$) were significantly associated with severe OSA. These findings suggest a cutoff BMI for predicting severe OSA of greater than 29.2 kg/m² with 81.3% sensitivity and 48.5% specificity. Severe OSA is common in children with obesity; thus, we recommend screening children with obesity and a BMI greater than 29.2 kg/m² for severe OSA.

Is obstructive sleep apnea a risk factor for diabetes?

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Obstructive sleep apnea (OSA) and type 2 diabetes are both closely related to obesity and their prevalence is increasing due to the rising average body weight in Western countries. The findings of epidemiological studies have implicated that OSA increases the risk for cardiovascular disease, and metabolic disturbances, such as insulin resistance, may link OSA to vascular morbidity. A number of observational clinical studies have evaluated the relationship between OSA and insulin resistance, suggesting an independent association. However, the confounding effect of obesity complicates the establishment of a causal relationship between OSA and insulin resistance. Potential mechanisms that may underpin this relationship were evaluated in animal and human experimental studies and include intermittent hypoxia, arousals from sleep with concomitant sympathetic activation and sleep fragmentation. Currently only three randomized controlled trials investigating the effects of OSA on insulin resistance have been published. In these trials OSA patients were randomly assigned to treatment with continuous positive airway pressure (CPAP) or subtherapeutic CPAP and treatment effects on various measures of insulin resistance were examined. In two of these trials there was no effect of CPAP on glucose metabolism and in one trial a small beneficial effect of CPAP was observed. Further carefully conducted clinical studies and randomized controlled interventional CPAP trials are needed to determine the extent to which OSA is a risk factor for diabetes and its effect on glucose metabolism.

Sleep apnea and obesity.

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The dramatic increase in the worldwide prevalence of obesity has paralleled the increase in the prevalence of obstructive sleep apnea (OSA). Even with heightened awareness by the lay and medical communities, OSA is still markedly under-diagnosed, as evidenced by the persistent presentation of late-stage cardiovascular complications in obese individuals newly diagnosed with sleep apnea. The clinical sequela of untreated and poorly-treated sleep apnea include conditions that are considered components of the metabolic syndrome for which central obesity is one of the major case-defining features. Hence, in this review of obesity and sleep apnea, it is unavoidable to include discussion of sleep apnea and other components of the metabolic syndrome. Proponents of this clinical perspective suggest that there are mutual genetic determinants that give rise to common phenotypic features and allow clustering of sleep apnea with the other components of the metabolic syndrome. Perhaps, the strongest observational evidence to support a link between sleep apnea and obesity is the similarity in age distribution of symptomatic sleep apnea and metabolic syndrome. The putative causal links between sleep apnea and each individual component of the metabolic syndrome have been extensively evaluated and have implicated bidirectional causality in certain metabolic conditions, such as obesity and sleep apnea, sleep apnea and diabetes mellitus, and obesity and diabetes mellitus. These studies collectively suggest that even modest weight loss improves OSA, and positively affects both metabolic and cardiovascular risk profiles.
